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3.4 General Design of the 10MW HTR

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Abstract:

According to the development programme of advanced technologies in China, a 10MW side-by-side high temperature gas-cooled test module reactor with spherical fuel elements will be erected before the year 2000 by the Institute of Nuclear Energy Technology of Tsinghua University. The aims of this project are to demonstrate the inherent safety features of the HTGR modular design and to test the technologies of electricity generation, district heating as well as process heat application with modular HTGR.

At present, the 10MW HTR main building is being constructed and the principle large components have been ordered and are being fabricated.

The design philosophy and safety features, the reactor structure, the principle components, the design parameters, the general layout of principle systems in the nuclear island of the 10MW high temperature gas-cooled test module will be presented in this paper.

General Design of the 10MW HTR

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Abstract

According to the plan of advanced technology development of our country, a 10MW High Temperature Gas cooled Test reactor (HTR-10) was designed by INET of Tsinghua University. At present, the reactor building is being constructed and the principal components are being fabricated or under goods. This paper describes the technical design of the HTR-10 test reactor.

1. General description

The Fig. 1 shows the HTR-10 primary system. The primary pressure boundary consists of reactor pressure vessel, steam generator pressure vessel and hot gas duct pressure vessel which connects the above two vessels. This arrangement can make the maintenance and inspection of the components easier and mitigate the accident result of water ingress into reactor core if the steam generator tubes were broken. We hope that the inherent safety features of side-by-side style HTR module can be demonstrated.

2. Reactor

The reactor core volume is 5m^3 , 1.8m in diameter and the mean height is 1.97m. About 27,000 spherical fuel elements with 60mm in diameter will be filled up in the reactor core, the enrichment of fuel is 17% and the mean burn up is 80,000MWd/T. At the funnel bottom of the reactor core, there is a fuel elements discharging tube, about 3.3m long and 500mm in diameter, at its end part through a special discharge facility the tube becomes a smaller one with 65mm in diameter only and penetrates the reactor pressure vessel. The fuel elements would be discharge one by one into the discharge system during the normal operation condition, therefor the shielding and cooling of the discharge tube will be easy. At the top of reactor core there is one loading tube, the internal diameter is 62mm.

Around the reactor core there are graphite reflectors, they can be divided into three parts: top reflector, side reflector and bottom reflector. In the outside of graphite reflector, there are carbon bricks containing thermal neutron absorber materials (B_4C), therefore they have thermal insulation and absorbing functions.

The whole graphite reflector consists of 20 layers of graphite bricks and carbon bricks. Each layer is divided into 20 segments, the graphite key and reactor core sleeve put them all together.

In the side reflector near the core there are 10 bore holes for control rods, the 10 control rods serve as the first shutdown system of HTR-10. According to our safety authority requirement there are 7 bore holes for small boron absorber balls as independent second shutdown system.

The reactor pressure vessel is more than 11m height and more than 4m in diameter, the reactor total weight is more than 300t.

3. Steam generator

The steam generator is once through modular helical tubes style. There are 30 small helical tubes modules which are installed in the annular space, each tube is 34m long, the total heat transfer area is $56m^2$ (See Fig. 2). The feed water temperature is $104^\circ C$, the temperature of outlet steam is $440^\circ C$, the steam pressure is 4MPa. At the center, a N_2 -He intermediate heat exchanger (IHX) will be installed in the second experimental stage for gas-turbine and process heat application experiment and in this case the helium inlet and outlet temperature would be from first stage $250^\circ C$ and $700^\circ C$ increased to $300^\circ C$ and $900^\circ C$ respectively, but at the first stage the place for the IHX is empty. The pressure vessel is more than 11m height and 2.6m in diameter.

4. Hot gas duct

The hot gas duct pressure vessel connects the reactor and steam generator pressure vessels, the center tube with 300mm in diameter, the hot helium will flow from reactor core to steam generator through this tube, so between the inside tube and outside tube a thermal insulation will

be installed. The outside tube with 900mm in diameter, the annular space between those two tubes is for cold helium flow from helium circulator into the reactor core, so the inside tube operation under high temperature but low pressure conditions.

5. The flow of coolant

The flow of the primary helium coolant is as following: After through the helium circulator, the pressure of the cold helium would be increased about 0.6 bar and flow through the annular space between the steam generator pressure vessel and the sleeve, that location in the outside of outside thermal insulation, and then come into the annular tube of the hot gas duct and then through the space between the reactor pressure vessel and the reactor core sleeve into the bottom part of the reactor support structure, the reactor support structure will be cooled at first, about 2% of cold coolant will be flowed into the control rods holes and 1% into the outside of fuel elements discharge tube for cooling, most of them will be risen through the 20 cold Helium bore holes enters the top cold helium plenum, after through the pebble bed, the cold helium was heated and from the hot Helium bore holes flow down to the hot Helium plenum, after mixture the average temperature will be 700℃, then through the central tube of the hot gas duct into the steam generator. After heat exchange the helium is cooled down from 700℃ to 250℃, then through 6 tubes enters the inlet of the helium circulator to form a closed circulation (See Fig. 3).

6. HTR-10 Confinement

Because the ceramic coated particles of fuel are selected in HTR-10, in our case under the normal operation condition and all accident conditions, the maximum temperature of the fuel elements are much more lower than the 1600℃ maximum temperature limit, so the coated particles are a strong barrier against release of radioactivity, and the HTR-10 has a large negative reactivity temperature coefficient of both fuel and moderator, its value is about $-8.5 \times 10^{-5}/K$. So under the normal operation condition, the maximum fuel elements temperature has a 700℃ margin to the limits of 1600℃. China nuclear safety authority agrees that for HTR-10 the containment is unnecessary, but we have a confinement design. The primary cavity is a principal part of confinement and the helium purification cavity is another part of confinement, therefor the primary system and all helium systems which are connected with primary system are covered by the confinement.

The primary cavity includes reactor cavity, steam generator cavity and fuel elements discharge system cavity. The wall thickness of reactor cavity is 2.3m mainly for shielding purpose. The design pressure of confinement is 1.3bar, during normal operation condition, the primary cavity is kept -150Pa lower other parts of reactor building through negative pressure system of the ventilation system, therefore the outside atmosphere can be released into the cavity, but the cavity inside gas couldn't release to the outside and only can be exhausted over 40m height chimney of ventilation center after filter.

If the primary lost pressure accident happened, for example, the helium purification system inlet and outlet tube or fuel elements discharge tube with 65mm in diameter one was damaged, the primary coolant will release very quickly and the pressure of primary cavity will be increased, when the pressure increased to 1.1 bar the flap or blast membrane will be opened, the gas will be exhausted form special chimney directly into atmosphere very quickly without filter (See Fig. 4).

In the reactor cavity a passive residual heat removal system is designed. It consists of reactor cavity coolers, air coolers and natural circulation loop. At normal shutdown condition the helium circulator and the steam generator can be operated, therefor the residual heat can be removed through the steam generator to the starting and shutdown loop, then to atmosphere. If above system was failed and at the starting and shutdown loop, then to atmosphere. If above system was failed and at the same time de pressurization accident was happened, the residual heat can be removed passively by means of conductivity of graphite reflector, reactor pressure vessel and then by means of radiation to the reactor cavity cooler, the residual heat will be removed passively over natural circulation, circulation loop, air cooler to the atmosphere.

Now the HTR-10 reactor building is being constructed. It consists of 6 floors, 3 of them are underground, the total height is 43m, the construction area is 6000 m². All 23 Nuclear inland systems will be installed in this building. The Fig. 5 and Fig. 6 show an example of systems layout. The Table 1 shows the general design parameters of HTR-10.

Table 1 General Design Parameters of HTR-10

1. Reactor Thermal Power	10MW
2. Design Life Time	20 years
3. Average Load Factor	50%
4. Reactor Core Volume	5 m ³
5. Reactor Core Diameter	180 cm
6. Equivalent Core Height at Equilibrium State	197cm
7. Fuel	UO ₂
8. Fresh Fuel U-235 Enrichment	17%
9. Diameter of Fuel element	6 cm
10. Heavy Metal Content of Each Ball	5 g
11. Fuel Loading Mode	multi-circulation (5 times)
12. Position of Control rods	side reflector
13. Control Rod Channels	10
14. Absorbing Ball Channels	7
15. Irradiation and Measuring Channels	3
16. Helium Inlet Temperature of Core	250℃
17. Helium Outlet Temperature of Core	700℃
18. Helium Mass Flow at 100% Power	4.3 kg/s
19. Primary Helium Pressure	3.0 MPa
20. Feed Water Temperature of SG	104℃
21. Feed Water Pressure of SG	6.1 MPa
22. Steam Outlet Temperature	440 ℃
23. Steam Outlet Pressure	4.0 MPa
24. Feed Water/Steam Mass Flow	3.49 kg/s
25. Residual Heat Removal Mode	natural circulation
26. Power of Residual Heat Removal System	250 kW

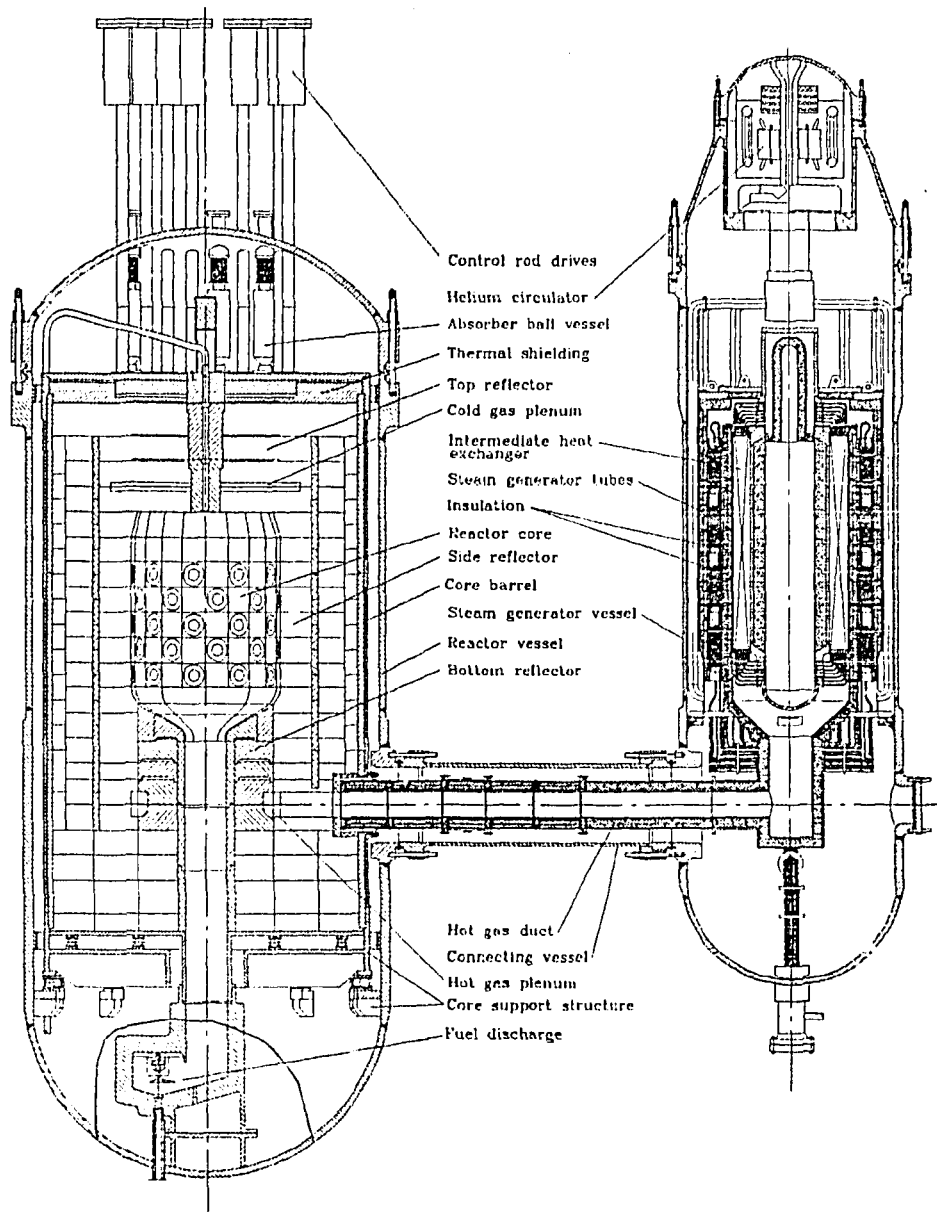


Fig.1 HTR-10 Primary System

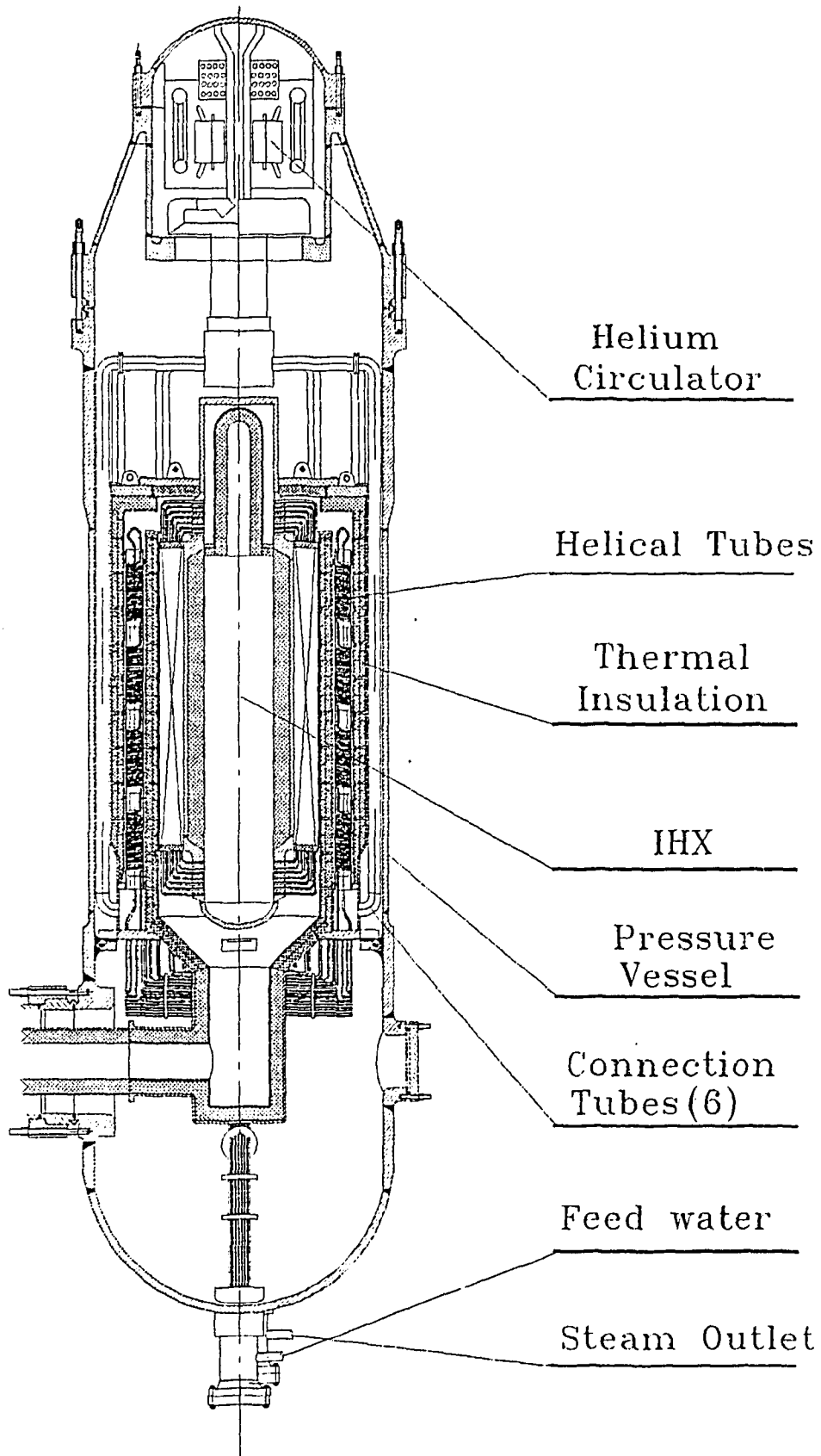


Fig. 2 HTR-10 Steam Generator

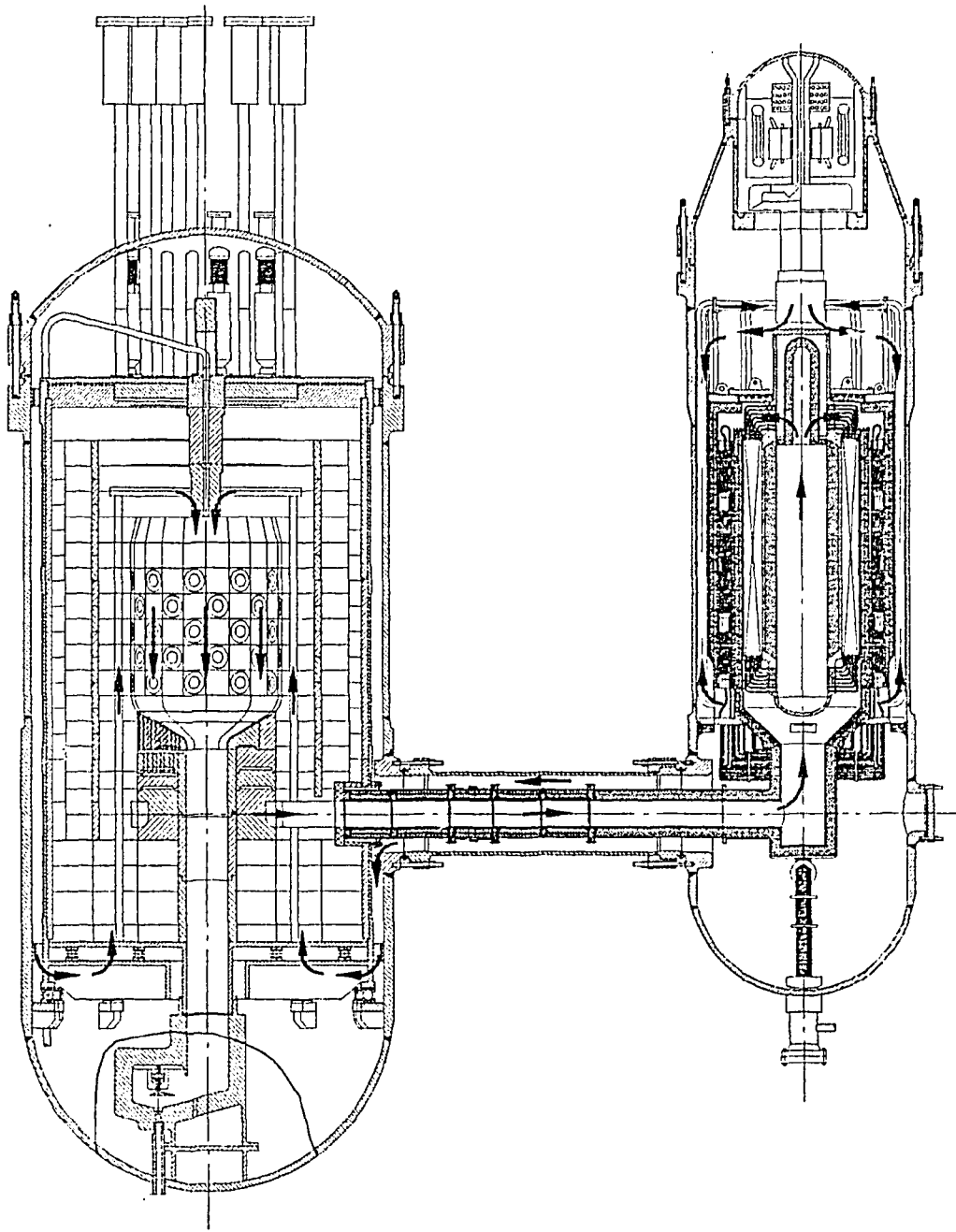


Fig3 The coolant flow of HTR-10

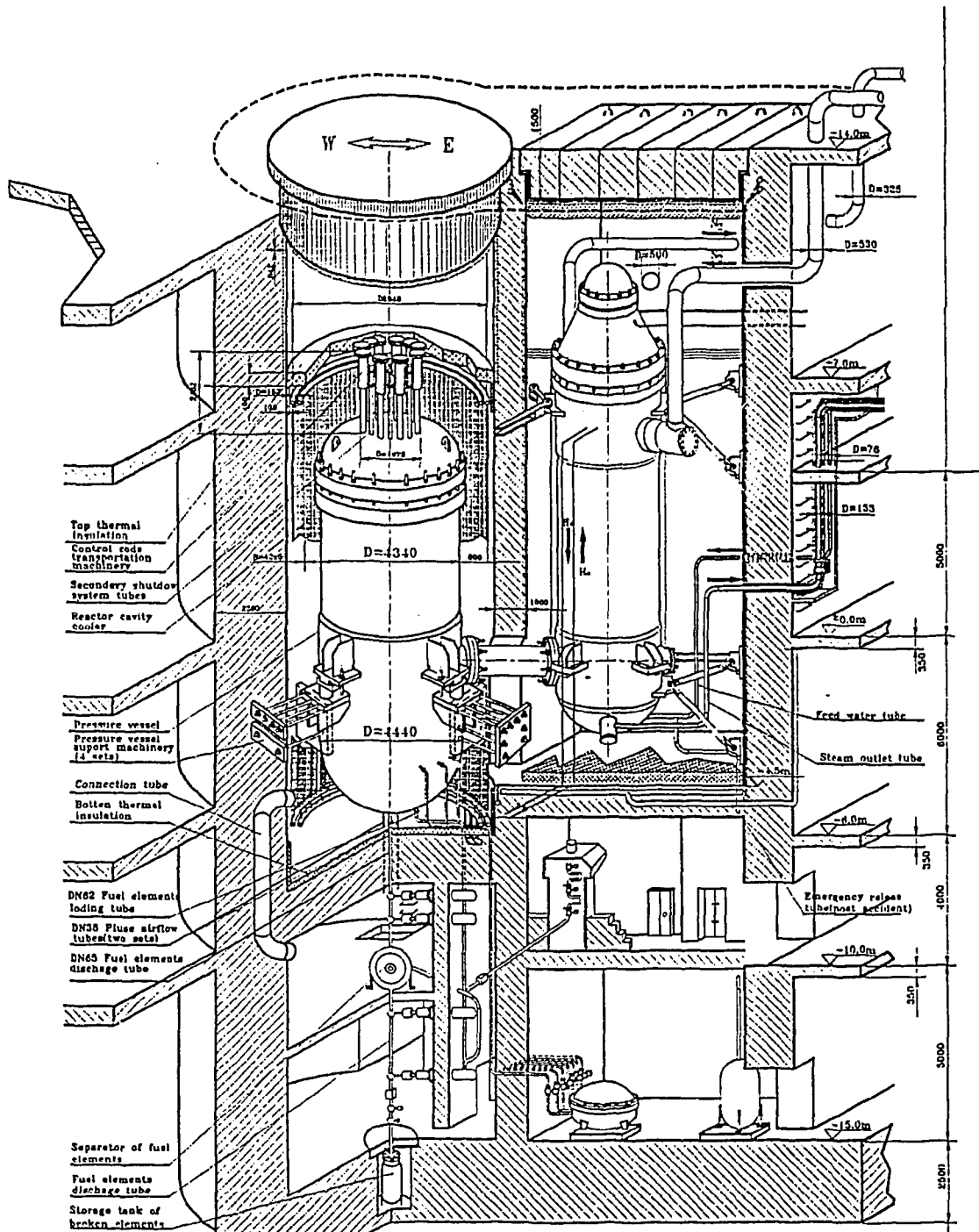


Fig.4 THR-10 Confinement

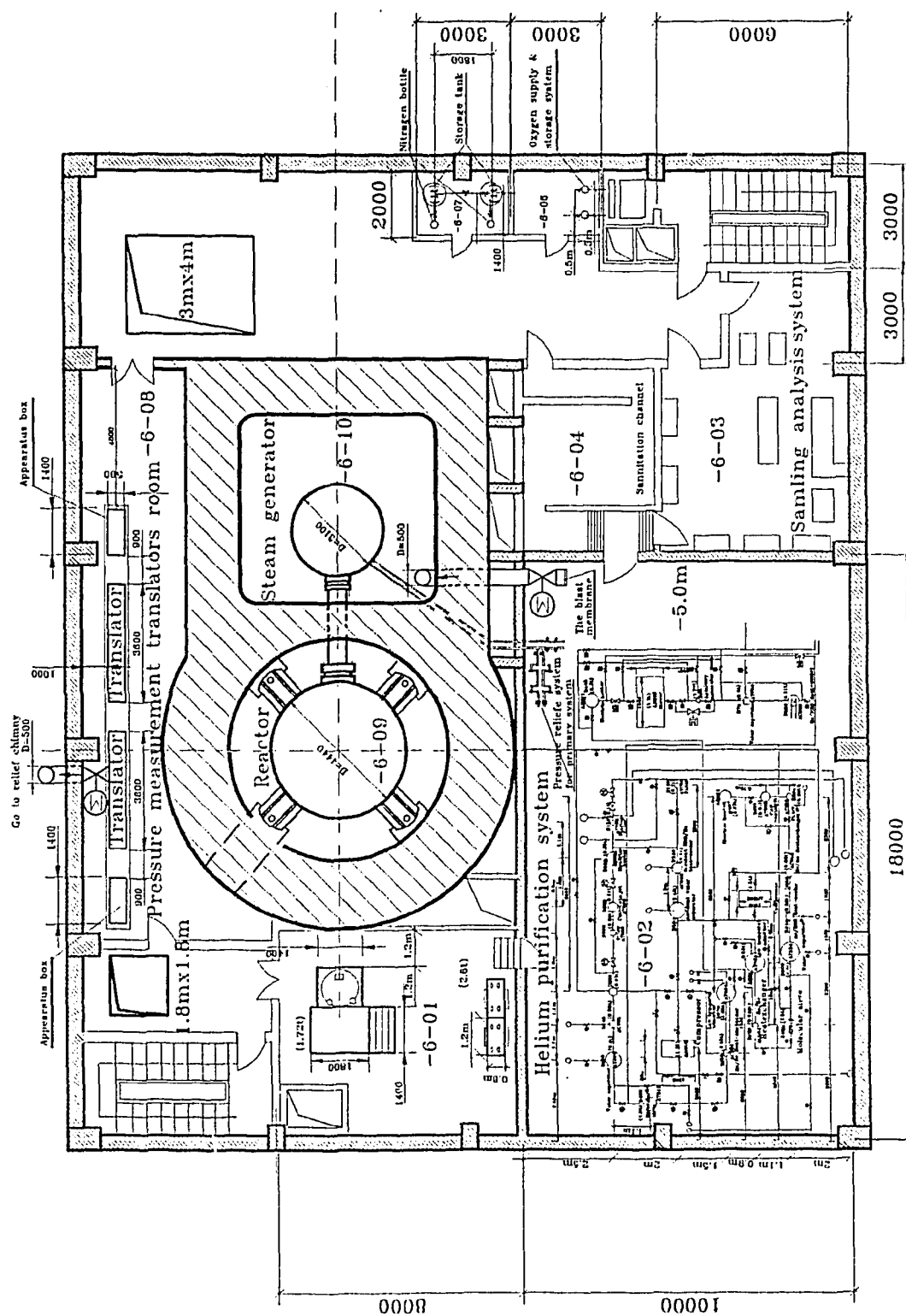


Fig.5 HTR-10 Systems Layout in -6.0m Floor

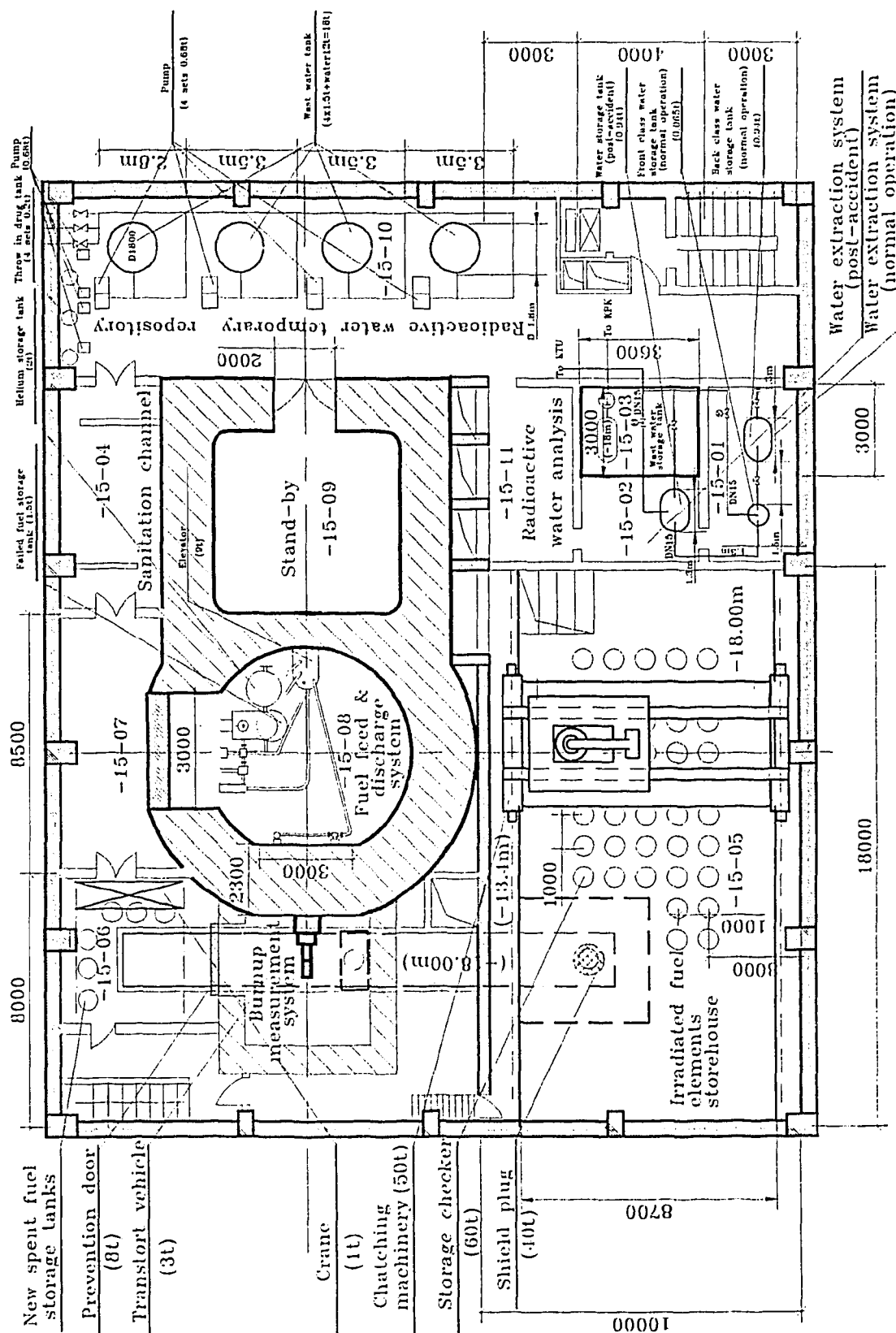


Fig. 6 HTR-10 Systems Layout in -15.0m Floor